

1. What are components of signal transduction?

The components of signal transduction include:

1. Signaling molecule (e.g., hormone)

2. Receptor: that specifically binds the signaling molecule;
Intracellular/nuclear receptors

G protein-coupled receptors Enzyme linked Receptor Ion channel-linked receptors

3. Transducer: Molecules that relay signals; G proteins

4. Second Messengers:

Small molecules that amplify the signal. examples: CAMP, CGMP, IP3, DAG, Ca²⁺

5. Effector Proteins: eg, kinase

6. Target Proteins: These include:

Transcription factors, Metabolic enzymes

7. Cellular Response: The outcome of signal transduction, such as gene expression, activation or inhibition of enzyme

2. Outline the steps of signal transduction mechanism.

3. Write a brief account of cAMP.

CAMP (cyclic adenosine monophosphate) is a second messenger of a certain water soluble hormone such as glucagon. These hormones bind the G protein coupled cell surface receptor. Binding of hormone to receptor activate G protein which in turn activate adenylate cyclase thereby producing CAMP. Binding of 4 cAMP to R subunits of protein kinase A causes dissociation of R₂ and C₂ resulting in activation of C subunits. Activated protein kinase A phosphorylates effector protein. Phosphorylation changes the activity of protein and so changes the metabolism.

Phosphatase, phosphodiesterase and intrinsic GTPase terminate the action of hormone.

4. Enumerate different types of hormones receptors with their respective ligands.

(1) Steroid hormone

- (i) Glucocorticoid
- (ii) Mineralocorticoid
- (iii) Sex hormone (E, M)
- (iv) Vitamin D

(2) Thyroid hormone (T₃, T₄)

5. List the hormones that bind to intracellular receptor.

Regulatory mechanisms

6. Mention the steps involved in biosynthesis of thyroid hormones and site of actions of antithyroid drugs.

- a. Concentration of Iodide (I^-)
- b. Oxidation of Iodide
- c. Iodination of tyrosyl residues
- d. Coupling of iodotyrosyls
- e. Secretion

Thiourea drugs can inhibit oxidation of I and coupling of iodotyrosyls.

7. List the steps of catecholamine biosynthesis.

1. Ring hydroxylation
2. Decarboxylation
3. Side chain hydroxylation
4. N-methylation
5. Storage and release of catecholamines

8. List the hormones that regulate blood calcium level.

Hormones that regulate calcium metabolism are

- (1) Parathyroid hormone (PTH)
- (2) Calcitriol ($1, 25 (OH)_2 D_3$)
- (3) Calcitonin (CT)

9. Write a brief account of parathyroid hormone.

Chemical nature - Protein.

Biosynthesis-PTH is synthesized as prepro PTH

Regulation of synthesis

PTH synthesis can be enhanced hypocalcemia or 1, 25 DHCC deficiency.

Regulation of secretion

PTH secretion is inversely related to concentration of ionized calcium level.

Mechanism of action

PTH binds to a single membrane receptor and initiates a typical cascade activation of adenylate cyclase and increasing cAMP level.

Action of parathyroid hormone

- PTH has a major effect on calcium homeostasis.
- It restored normal ECF calcium concentration by acting directly on bone and kidney and by acting indirectly on the intestinal mucosa.

Effect on bones

- Receptors of PTH are found on preosteoblasts, osteoblasts, lining cells and osteocytes. The most obvious effects of PTH are to increase bone formation.
- PTH activates osteoclasts indirectly via osteoblasts.
- When plasma calcium decreases. PTH stimulating osteoclasts-mediated bone reabsorption.

Effect on kidneys

- PTH decreases reabsorption of phosphate. This is known as phosphaturic action.
- PTH also increases reabsorption of Ca^{2+}
- PTH also increases the formation of 1,25-DHCC .

1	Fat (10%)	
	Lean body mass (30%)	
	Lymph (1.5L) (2.2%)	Intracellular fluid (40%)
	Slowly exchanging fluid (7.5L) (10.11%)	
	Transcellular fluid (1.5L) (2.2%)	
ISF (15%) (10.5L)		

1.

* Distribution of TBW

Total body weight of normal adult = 70kg

The total body water (TBW) - 60% of body weight (42L)

(1) Intracellular fluid (ICF): - 40% of body weight (28L)

(2) Extracellular fluid (ECF): - 20% of body weight (14L)

(2.1) Interstitial fluid (ISF): - 15% of body weight (10.5L)

i. Lymph, 2.2% (1.5L)

ii. Slowly exchanging fluids, 10-11% (7.5L)

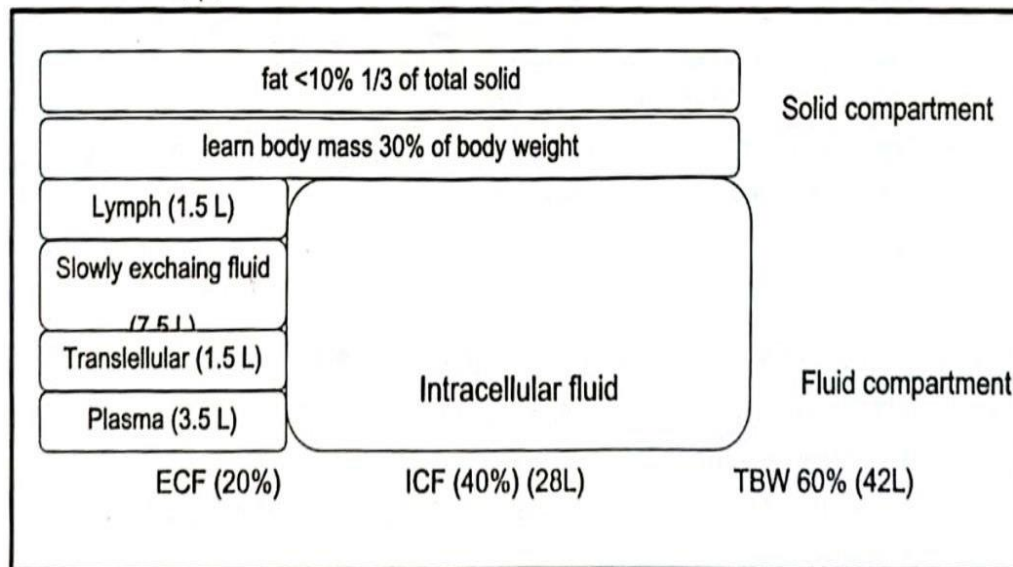
⇒ (bone, cartilage, connective tissue)

iii. Transcellular fluid (TCF), 2.2% (1.5L)

⇒ (fluids within the eye, alimentary secretions, cerebrospinal fluid, and pleural fluid, pericardial fluid, peritoneal fluid, synovial fluid, etc.)

(2.2) Intravascular fluid (IVF): - 5% of body weight (3.5L)

* Body Fluid Compartments



2.

Water

- (1) Acts as universal solvent.
- (2) Helps in thermoregulation (because of its high heat capacity and high latent heat of vapourization)
- (3) Protects against freezing (because of its high latent heat of solidification)
- (4) Influences the state of dissociation of macromolecules
- (5) Medium for transport of nutrients, gases etc.
- (6) Site where biochemical reactions take place.
- (7) Lubrication function (in swallowing, joint space)

3.

II. Electrolytes

- (1) Help in distribution of water (osmosis)
- (2) Exerts osmotic pressure, especially sodium, since over 90% of the osmotic activity of ECF is due to sodium present in the form of NaCl.

$$\text{Total body osmolarity} = \frac{\text{Total Na}^{\circ} + \text{total K}^{\circ}}{\text{TBW}}$$

exchangeable
TBW Total body water

('e') is the exchangeable form)

$$\text{Plasma osmolarity (P}_{\text{osm}}) = 2 (\text{Na}_{\text{mEq/L}}) + [\text{urea}] + [\text{Glucose}]$$

4. Composition of body fluids

1. Suspended colloids: - non-diffusible proteins like albumin
2. Crystalloids: - glucose, urea, uric acid, and other organic molecules such as lactate, acetate.
3. Electrolytes: - Na^+ , K^+ , Ca^{2+} , Mg^{2+} , Cl^- , HCO_3^- , PO_4^{2-} , SO_4^{2-}

5. Regulatory mechanisms

- (a) ADH (Antidiuretic hormone or Arginine vasopressin, AVP),
- (b) Thirst mechanism
- (c) Renin-Angiotensin-Aldosterone (RAA) System
- (d) Atrial natriuretic peptide (ANP) or Atriopeptin

6. Action

1. It causes arteriolar constriction and a rise in systolic and diastolic blood pressure.
2. It acts directly on the adrenal cortex to increase the synthesis and secretion of aldosterone. The aldosterone causes Na^+ and water retention, expanding ECF volume and shunting off the stimulus that initiated increased renin secretion.
3. It acts on the brain to stimulate the neural areas concerned with thirst to increase the water intake.
4. On posterior pituitary, it stimulates ADH secretion.
5. It facilitates the release of norepinephrine from postganglionic sympathetic neurons, further facilitating vasoconstriction.

6. It has direct effect on the renal tubules to increase Na reabsorption and messengial cells contraction. This tends to reduce glomerular filtration

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